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🌐 <https://scholar.google.com/citations?user=hCvIj5cAAAAJ&hl=en&oi=ao>



🎓 EDUCATION

University of Science and Technology of China & Shanghai AI Lab 📍 Hefei & Shanghai

Information and Communication Engineering, Ph.D. Candidate

2024.7 ~ 2027.6 (Expected)

- **Research Focus:** Multimodal understanding and generation, embodied intelligence world models, unified models, large language models, quantitative research
- **Advisors:** Prof. Feng Wu (CAE Academician, IEEE Fellow), Prof. Zhiwei Xiong; Co-supervisor: Prof. Xiaoou Tang (IEEE Fellow, Shanghai AI Lab)
- **Core Courses:** Algorithm Design and Analysis, Statistical Learning, Deep Learning, Reinforcement Learning
- **Honors:** Principal Investigator of National Natural Science Foundation Ph.D. Project (2024)

University of Science and Technology of China 📍 Hefei

Computer Technology, M.S.

2022.9 ~ 2024.7

- **Honors:** National Graduate Scholarship (2022)

Xiamen University 📍 Xiamen

Environmental Ecological Engineering & Economics (Dual Degree), B.S.

2018.9 ~ 2022.7

- **Overall Ranking:** 1/31
- **Honors:** Xiamen University Academic Star (2021), CDA Level 1 Certification (2022), Kaggle Expert
- **Advisor:** Prof. Yuanye Zhang

🔬 RESEARCH PROJECTS

Pelican-Unified 1.0: Unified Embodied Intelligence Model

2026.5

Technical Report Core Contributor (Ranked Second)

- Contributed to a unified embodied intelligence model for **understanding, reasoning, imagination, and action**.
- Helped build a shared framework that integrates multimodal understanding, chain-of-thought reasoning, future video rollout, and low-level action generation.
- Paper: arXiv PDF
- Media coverage: People's Daily, Beijing Daily / Beijing Gov, Machine Heart, Zhidongxi

National Natural Science Foundation Ph.D. Project

2025.1 ~ 2027.12

NSF Project Principal Investigator (Funding: **300k RMB**, **Only recipient** in Information field, Anhui Province, 2024)

[1] Foundation **10-billion-scale** brain neural **foundation model** training

- Neuron underlying visual feature **token construction**
- Neural architecture search, **autoregressive visual model** training

[2] **Fine-grained** neuron segmentation

- **Skeleton features**, attention decomposition mechanisms
- Complementary masking, segmentation **robustness enhancement**

[3] Neural **circuit reconstruction**

- Point cloud morphological features and image feature **alignment**
- Candidate competition strategy, **progressive reasoning**, long-range tracking

Large-scale Self-supervised Pretraining

2022.5 ~ 2023.12

Conference&Journal First Author

[1] **[LONG ORAL]** Self-supervised neuron segmentation with multi-agent reinforcement learning, IJCAI 23 **[CCF-A]**

- Improved MAE masking strategy using reinforcement learning, automatically selecting mask ratios and schemes.
- Introduced multi-agent framework for more efficient self-supervised feature learning, improving segmentation accuracy by 12%.

[2] **MaskTwins: Dual-form Complementary Masking for Domain-Adaptive Image Segmentation**, ICML 25 **[CCF-A]**

- Proposed new complementary masking theory from sparse signal reconstruction perspective, rigorously proving theoretical advantages of dual-form complementary masking in extracting domain-invariant features.
- Designed simple and efficient unsupervised domain adaptation framework, achieving cross-domain image segmentation through complementary masking consistency learning, improving natural image segmentation by 2.7% mIoU and biomedical image segmentation by 3.2% IoU.

[3] **TokenUnify: Scalable Autoregressive Visual Pre-training with Mixture Token Prediction**, ICCV 25 **[CCF-A]**

- Proposed image autoregressive pretraining combined with Mamba framework, demonstrating advantages of long sequences and low computational cost.

- Demonstrated good scaling laws with corresponding theoretical proof, showing logarithmic-linear relationship between model performance and parameters.

[4] EMPOWER: Evolutionary Medical Prompt Optimization With Reinforcement Learning, JBHI [SCI-Q1]

- Proposed the first evolutionary prompt optimization framework for medical domain, combining domain knowledge and reinforcement learning.
- Core techniques: medical terminology attention mechanism, multi-dimensional quality assessment (clarity/specificity/relevance/accuracy), structure-preserving evolutionary algorithm, semantic verification module ensuring clinical guideline consistency.
- Reduced factual errors by 24.7%, improved domain specificity by 19.6%, and achieved 15.3% higher clinician preference in blind testing.

[5] Learning multiscale consistency for self-supervised electron microscopy instance segmentation, ICASSP 24 [CCF-B]

- Achieved high-performance pretraining strategy based on multiscale feature contrastive learning and feature reconstruction.
- Proposed feature consistency loss function, overcoming representation differences caused by scale variations, improving accuracy by 9%.

[6] [Major Revision] Generative Text-Guided 3D Vision-Language Pretraining for Unified Medical Image Segmentation, Submit to PR

- Generated image descriptions using large language models for multimodal image-text contrastive learning pretraining.
- Innovatively introduced generative text guidance mechanism, solving medical image annotation scarcity problem, achieving zero-shot segmentation.

Large-scale Data Generation and Encoding Compression

2023.8 ~ Present

Conference&Journal First Author & Co-first Author

[1] [Accepted] Learned Image Coding with Generative Reference of Conditional Latents, TPAMI [SCI-Q1]

- Extension work of AAAI 25 oral paper, further exploring benefits of reference images for image coding.
- Obtained reference images through local dictionary, network retrieval, and image generation, improving compression performance by 15%.
- Proposed conditional latent generation framework, effectively solving performance degradation when reference images are unavailable.

[2] [ORAL] Conditional Latent Coding with Learnable Synthesized Reference for Deep Image Compression, AAAI 25 [CCF-A]

- Built image similarity dictionary for retrieving similar images to improve entropy model probability estimation.
- Proposed learnable synthetic reference framework, improving PSNR by 0.6dB at the same bit rate with leading compression performance.

[3] MaskFactory: Towards High-quality Synthetic Data Generation For Dichotomous Image Segmentation, NeurIPS 24 [CCF-A]

- Generated corresponding mask-image pairs using ControlNet through rigid and non-rigid deformation editing of masks.
- Core techniques: deformation-driven mask editing, ControlNet-guided image generation, synthetic data quality assessment.
- Synthetic data performance approached real data in downstream segmentation tasks with only 2% performance gap, significantly reducing annotation costs.

[4] [Under Review] Joint Semantic and Coded Generation for Conditional Latent Coding, Submit to TCSVT

- Proposed joint semantic and coded generation framework, combining text descriptions with minimal coded features (0.01 bpp) to guide controllable rectified flow models in generating high-fidelity reference images.
- Core techniques: ControlNet for DiT architecture, iterative reference alignment (IRA), dynamic feature fusion (DFS), iterative probability refinement (IPR), theoretical robustness proof.
- Improved compression performance by 15.8% BD-rate compared to VVC, successfully unifying image generation and compression, bridging the gap between semantic and coded representations.

[5] BIMCV-R: A Landmark Dataset for 3D CT Text-Image Retrieval, MICCAI 24 [CCF-B]

- Built the first open-source 3D CT image-text pair dataset containing 10,000 high-quality medical image-description pairs.
- Achieved efficient image-text information retrieval and keyword search, improving recall rate by 25% over traditional methods.

[6] [Under Review] UniCompress: Enhancing Multi-Data Medical Image Compression with Knowledge Distillation, Submit to TCSVT

- Achieved implicit neural network compression of multiple data through multimodal knowledge priors, improving compression rate by 40%.
- Extracted common features of various medical images through knowledge distillation, reducing storage space by 20% while maintaining diagnostic quality.

Large Model Engineering

2023.9 ~ Present

Projects Core Member

[1] Image coding and intra-frame prediction large models

- Led design of billion-parameter-scale coding architecture, improving compression rate by 30% over traditional coding standards.

[2] Medical image segmentation and neuron segmentation large models

- Mainly responsible for pretraining in the team, with experience in large-scale cluster pretraining using 64 A40 GPUs.

- Proficient in large model frameworks like DDP and DeepSpeed, achieving efficient training and optimization of 30-billion parameter models.

🧑‍🔬 RESEARCH EXPERIENCE

Ubiquant	📍 Beijing	2026.5 ~ Present
Quantitative Research and Large Language Models <i>Researcher</i>		
• Working on quantitative research and large language model research. • Exploring how multimodal and language-model research can support data-driven decision making.		
Beijing Humanoid Robot Innovation Center (UBTECH Robotics)	📍 Beijing	2025.12 ~ Present
Embodied Intelligence World Model Algorithm Team <i>Core Member</i>		
• Focusing on embodied intelligence world models and unified models for humanoid robots. • Researching unified multimodal perception and decision-making frameworks for humanoid robots. • Developing world model-based robot behavior prediction and planning algorithms. • Contributed as a core contributor (ranked second) to Pelican-Unified 1.0, a technical report on unified embodied intelligence.		
Tencent (IEG)	📍 Shanghai	2025.8 ~ 2025.12
Qingyun Program <i>Research Intern</i>		
• Focused on game video scene understanding technology development, building intelligent video content analysis systems. • Designed game scene automatic recognition and classification algorithms based on multimodal large models, achieving accuracy above 95%. • Trained TTS models for Honor of Kings Lingbao (based on f5-tts and indextts2).		
People's Liberation Army General Hospital (301 Hospital)	📍 Beijing	2023.9 ~ 2024.2
Data Compression Group <i>Research Intern</i>		
• Collaborated with Academician Qionghai Dai's team on efficient data compression research. • Designed medical imaging-specific compression algorithms optimized for CT, MRI modalities, improving compression efficiency by 35%.		
Imperial College London	📍 London (Remote)	2022.11 ~ 2023.8
Data Science Institute <i>Research Intern</i>		
• Collaborated with Associate Professor Rossella Arcucci on multimodal pretraining research, submitting one journal paper. • Developed image-text contrastive learning framework, achieving 93.5% accuracy on medical diagnosis tasks.		
Xiamen University WISER Club	📍 Xiamen	2021.8 ~ 2022.7
Data Mining Group <i>Insider</i>		
• Responsible for designing and discussing data mining courses, teaching clustering and Transformer sections. • Mentored 20 undergraduate students to complete machine learning projects, organized 2 campus competition activities.		
Wang Yanan Institute for Studies in Economics, Xiamen University	📍 Xiamen	2020.8 ~ 2021.12
Econometrics <i>Research Assistant</i>		
• Assisted Associate Professor Jiong Zhu in completing territorial economic statistics, conducting visual feature extraction of homestead information. • Developed satellite image analysis tools for automatic land use change identification with 85% accuracy.		

🏆 HONORS AND AWARDS

• National Natural Science Foundation Ph.D. Project	2024.12
– Principal Investigator, Only recipient in Information field, Anhui Province	
• Interdisciplinary Contest in Modeling (ICM), Outstanding Winner	2024.05
– Top 0.17% of 10,388 participating teams worldwide	
• National Graduate Scholarship	2022.12
– Award rate 1%	
• Xiamen University Academic Star	2021.12
– Only undergraduate recipient	
• 'Jingrun Cup' Mathematics Competition Professional Group	2021.09
– First Place, Xiamen University	
• 'Internet+' Competition	2021.08
– Gold Award, Fujian Province	
• National College Student Mathematics Competition Non-major Group Finals	2021.05
– National Second Prize	
• 'Challenge Cup' National College Student Extracurricular Academic Science and Technology Competition	2021.05
– First Prize, Fujian Province	
• National College Student Mathematics Competition Non-major Group	2020.11
– First Place, Fujian Province	

🔧 TECHNICAL SKILLS

- **Programming:** Python, C, C++, Java, MATLAB, $\LaTeX$
- **Deep Learning:** PyTorch, TensorFlow, DeepSpeed, DDP
- **Languages:** English (TOEFL 110, GRE 328), Chinese (Native)
- **Tools:** Git, Docker, CUDA, HPC

 PROFESSIONAL SERVICE

- **Reviewer:** CVPR 2025, ICCV 2025, WACV 2026, NeurIPS 2024, ICML 2025, ICLR 2024, MICCAI 2025, ACM MM 2024, AISTATS 2024, IJCV, TIP

 RECENT NEWS

• Joined Ubiquant for quantitative research and large language model research	2026.05.26
• Contributed to Pelican-Unified 1.0 as a core contributor (ranked second)	2026.05.15
• Two papers were accepted by ICLR 2026	2026.01.26
• One paper on medical image registration was accepted by ICASSP 2026	2026.01.18
• Joined Beijing Humanoid Robot Innovation Center (UBTECH Robotics) as core member of Embodied Intelligence World Model Algorithm Team	2025.12
• Advisor Prof. Feng Wu was elected as Academician of the Chinese Academy of Engineering (CAE)	2025.11.21
• One paper was accepted by AAAI 2026	2025.11.09
• One paper was accepted by TCSVT	2025.11.05
• One paper was accepted by JBHI	2025.10.16
• One paper was accepted by ICCV 2025	2025.06.26
• One paper was accepted by ACL 2025 findings	2025.05.15
• One paper was accepted by ICML 2025	2025.05.01
• One paper was selected as oral by AAAI 2025	2025.01.18
• Successfully selected as principal investigator of Ph.D. Natural Science Foundation Project	2024.12.06
• One paper was accepted by NeurIPS 2024	2024.10.10